

Sakana Fugu evaluation

ProcessSmith · AI provider assessment · 27 June 2026

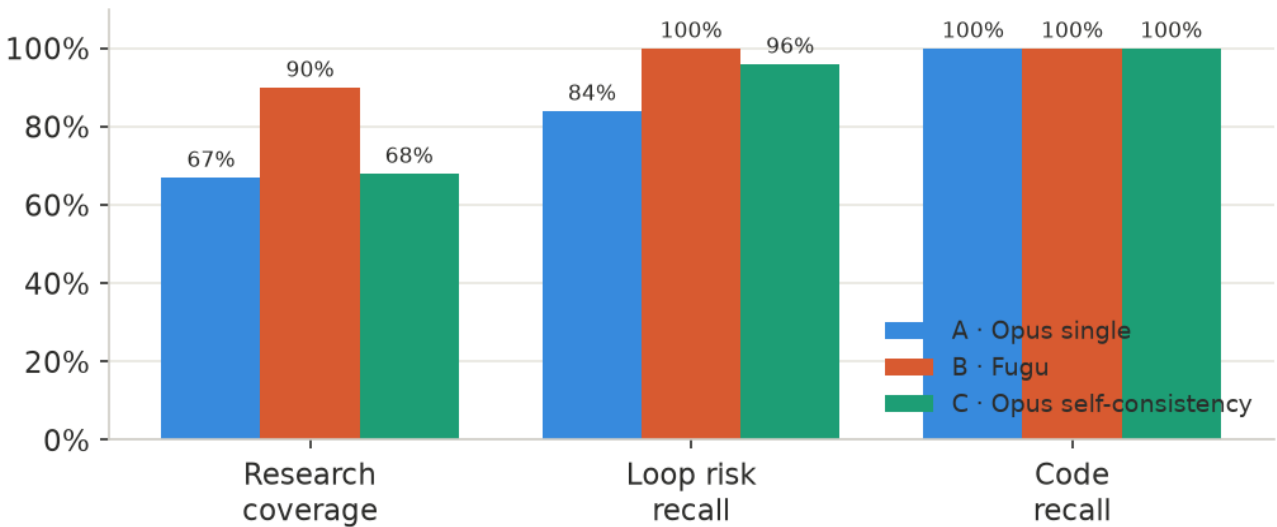
Decision: Trial — a niche hard-research / risk-critique backstop, run alongside the default Opus path. Not a default runtime; not for code review.

A pre-registered, three-arm evaluation tested Sakana’s fugu-ultra against a single Claude Opus 4.8 pass and an Opus self-consistency ensemble, across 15 frozen tasks in three classes (research briefs, code review, automation-risk critique), plus a five-item hard code-review tiebreaker. Across 90 model runs (~\$8 total, all completed clean), Fugu produced materially better, source-grounded output that **both** Opus arms missed in research and risk-critique — but at roughly 4× the cost and 10× the latency, with no advantage on code review. Hence Trial: a second-opinion backstop, never the default.

Total spend	Fugu vs Opus	Quality wins	Default runtime
\$8 90 runs, all clean	~4x cost ~10x slower	2 of 3 code = tie	Opus cost-adjusted champ

Arm	System	Cost / 15 items	Latency / item
A	Opus 4.8, single pass	\$0.56	22 s
B	fugu-ultra	\$2.48	216 s
C	Opus 4.8 x3 + synthesis	\$2.71	84 s

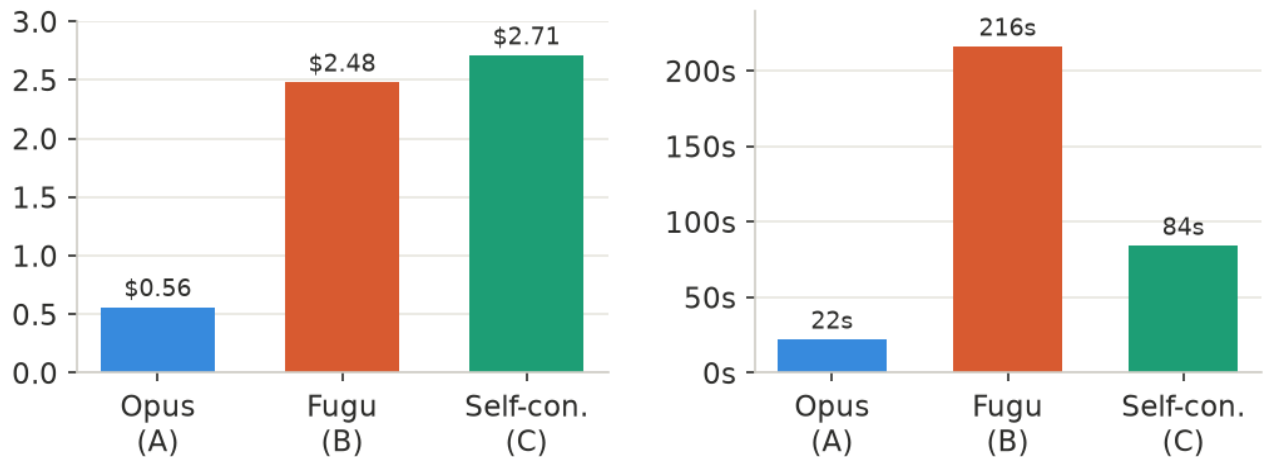
Quality by class



- **Research** — Fugu led at ~90% coverage with real, independently-verified citations and no fabrication; both Opus arms refused the knowable facts in one question, costing them ~8 points each.

- **Loop-critique** — Fugu caught every seeded risk (25/25) and surfaced the richest unique risks (control-plane / data-plane separation, TOCTOU approval-binding, a conservation invariant). Self-consistency 24/25; single Opus 21/25.
- **Code review** — a three-way tie: all three arms found every planted defect.

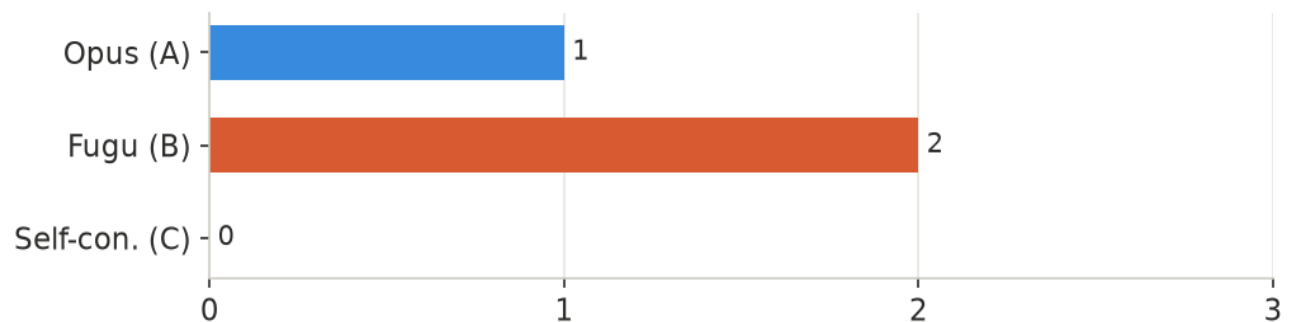
Cost and latency — the case against default use



Fugu is the tall bar in both: ~4× the spend and ~10× the wall-clock of a single Opus pass. A marginal quality edge that runs 10× slower is a backstop, not a default.

The twist — code-review precision on hard bugs

A harder tiebreaker set (expertise-level bugs: re-serialized-body signature confusion, a lock-that-doesn't-lock, penny-allocation). All three arms still found 10/10 — but they split on false flags raised against *correct* code (lower is better):



Fugu raised the most false flags — its extra deliberation tipped into over-flagging. Code review is firmly excluded from its use.

Recommendation

Use Fugu as a backstop for

- Hard research synthesis
- Adversarial risk / safety critique
- Second opinion where breadth matters

Don't use Fugu for

- Code review (ties on recall, over-flags)
- Latency-sensitive or interactive work
- Routine tasks (Opus is ~5x cheaper)
- Client-confidential data, or as the default

Caveats. n = 5 per class — a go/no-go signal, not proof. First-pass scoring was by independent scorers against frozen answer keys; a human re-score is recommended, especially for research citations. The code class ceilinged on recall in both rounds. Single Opus + adaptive thinking remained the cost-adjusted champion; self-consistency edged it only modestly at ~5x the price — reconfirming the Debate-Harness finding that one strong model is hard to beat.

Source — processsmith-research PR #2: full brief, evaluation plan, frozen pre-registrations, detailed scoresheet, and the reproducible harness are committed. ClickUp task 86bamrnvm.